

SSH Brute Force and Post-Exploitation Activity Detected via Honeypot

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1.Executive Summary

An SSH honeypot was deployed on a public cloud instance to monitor unauthorized access attempts. During the observation period, multiple brute-force login attempts were detected originating from external IP addresses. One or more attackers achieved simulated login success and executed post-compromise discovery commands and attempted malware downloads. All activities were captured and analyzed using Cowrie JSON logs.

2.Environment Details

- Cloud Provider: AWS EC2
- Operating System: Ubuntu 22.04 LTS
- Honeypot Technology: Cowrie SSH Honeypot
- Listening Port: 2222
- Log Source: cowrie.json

3.Detection Overview

- Detection Method: Network-exposed SSH honeypot
- Attack Types Observed:
 - Credential brute-force attacks
 - Unauthorized command execution
 - Simulated malware delivery attempts

4.Timeline of Events

Time (UTC)	Event Description
09:42:38	Successful SSH login detected using username <code>root</code> and password <code>admin</code>
09:42:39	Attacker session established (Session ID: <code>3ddac984b932</code>)
09:42:41	Post-authentication activity initiated by attacker
09:42:43	System discovery commands executed (e.g., <code>whoami</code> , <code>uname -a</code>)

Time (UTC)	Event Description
09:42:47	File system enumeration activity observed
09:42:52	Attempted external file retrieval indicative of malware staging
09:42:58	SSH session terminated by attacker

5.Evidence Collected

Log Source:

/var/log/cowrie/cowrie.json

Sample Evidence (Redacted):

```
{"eventid":"cowrie.login.success","username":"root","password":"admin","timestamp":"2025-12-30T09:42:38Z","src_ip":"103.70.197.99","session":"3ddac984b932"}
```

(The above log entry confirms successful authentication and was used to reconstruct the attacker session timeline.)

Screenshot 1 — Honeypot Log Evidence

(Figure 1 shows Cowrie honeypot logs capturing a successful SSH login attempt followed by post-authentication command execution. The log entries were used to reconstruct the attacker timeline and analyze behavior.)

```
cowrie@ip-172-31-32-138: ~$ cat /var/log/cowrie/cowrie.json
{"message":"SSH client hash fingerprint: 701158e75b580e76f0410d5d22ef9d40","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:42:27.683663Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.login.success","username":"root","password":"admin","message":"login attempt [root/admin] succeeded","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:42:38.462859Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.client.size","width":102,"height":83,"message":"Terminal Size: 102 53","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:42:38.916153Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.session.params","arch":"Linux-x86_64-lsb","message":"","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:42:38.917284Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.command.input","input":"whoami","message":"CMD: whoami","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:43:25.462305Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.command.input","input":"uname -a","message":"CMD: uname -a","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:43:34.743509Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.command.input","input":"ls","message":"CMD: ls","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:43:38.709808Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
{"eventid":"cowrie.command.input","input":"cd /tmp","message":"CMD: cd /tmp","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:43:45.169212Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
(cowrie-env) cowrie@ip-172-31-32-138:~/cowrie/var/log/cowrie$ grep <SESSION_ID> cowrie.json
-bash: SESSION_ID: No such file or directory
(cowrie-env) cowrie@ip-172-31-32-138:~/cowrie/var/log/cowrie$ cd ~/cowrie/var/log/cowrie
(cowrie-env) cowrie@ip-172-31-32-138:~/cowrie/var/log/cowrie$ grep login success cowrie.json
{"eventid":"cowrie.login.success","username":"root","password":"admin","message":"login attempt [root/admin] succeeded","sensor":"ip-172-31-32-138","uid":"5d25d32d-e563-11f0-a905-0ac8e5deaa07","timestamp":"2025-12-30T09:42:38.462859Z","src_ip":"103.70.197.99","session":"3ddac984b932","protocol":"ssh"}
(cowrie-env) cowrie@ip-172-31-32-138:~/cowrie/var/log/cowrie$
```

